



Configuring Multi-Area OSPF to Implement Dynamic Routing

Difficulty (out of 5 stars): ★★★

Recommended study time: 3 hours

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1 About This Lab

1.1 Task Description

An enterprise does not want to spend a large amount of manpower to maintain network performance, and requires that new routes be calculated based on route update packets. In this lab, deploy OSPF to ensure network stability and interconnection.

After completing this task, you will be able to independently set up and configure medium- and large-sized networks using OSPF.

1.2 Objectives

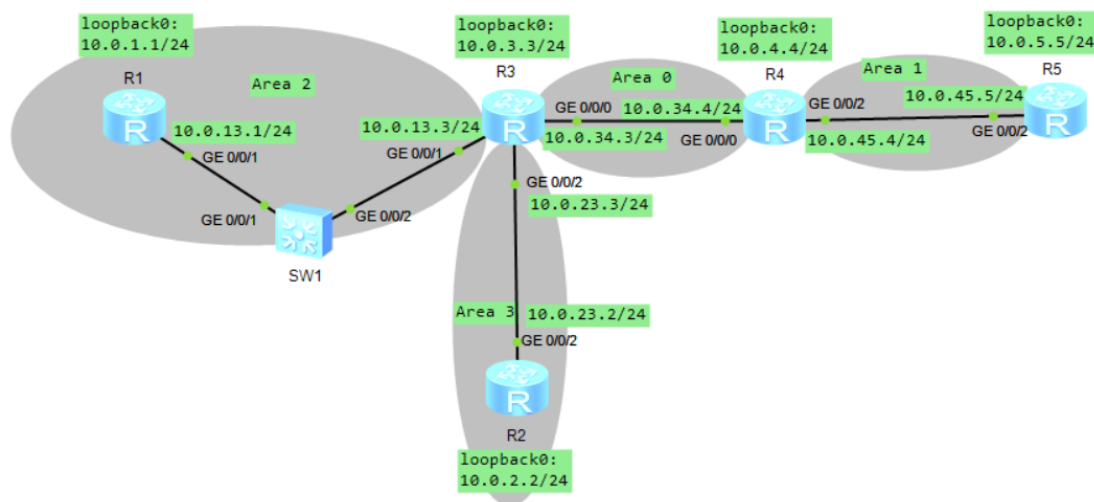
On completing this task, you will be able to:

1. Configure multi-area OSPF.
2. Configure inter-area route summation.
3. Enable a device to import external routes into an OSPF area.
4. Configure stub and totally stub areas.
5. Configure an NSSA.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology



The preceding figure shows the device interconnection mode and IP address plan. The OSPF areas are planned as follows:

1. The interconnection interfaces between R1 and R3 and R1's Loopback0 belong to OSPF area 2.
2. The interconnection interfaces between R3 and R4 and their Loopback0 interfaces belong to OSPF area 0.



3. The interconnection interfaces between R4 and R5 belong to OSPF area 1, and R5's Loopback0 does not belong to any area.
4. The interconnection interfaces between R2 and R3 belong to OSPF area 3, and R2's Loopback0 does not belong to any area.

2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int LoopBack 0
[R1-LoopBack0]ip address 10.0.1.1 24
[R1-LoopBack0]quit
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.0.13.1 24
[R1-GigabitEthernet0/0/1]quit
[R1]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int lo0
[R2-LoopBack0]ip address 10.0.2.2 24
[R2-LoopBack0]int g0/0/2
[R2-GigabitEthernet0/0/2]ip address 10.0.23.2 24
[R2-GigabitEthernet0/0/2]quit
[R2]
```

- Initial configuration of R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]int LoopBack 0
[R3-LoopBack0]ip address 10.0.3.3 24
[R3-LoopBack0]int g0/0/1
[R3-GigabitEthernet0/0/1]ip add
[R3-GigabitEthernet0/0/1]ip address 10.0.13.3 24
[R3-GigabitEthernet0/0/0]int g0/0/0
[R3-GigabitEthernet0/0/0]ip address 10.0.34.3 24
[R3-GigabitEthernet0/0/0]int g0/0/02
[R3-GigabitEthernet0/0/2]ip address 10.0.23.3 24
[R3-GigabitEthernet0/0/2]quit
[R3]
```

- Initial configuration of R4

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R4
```



```
[R4]int LoopBack 0
[R4-LoopBack0]ip address 10.0.4.4 24
[R4-LoopBack0]int g0/0/0
[R4-GigabitEthernet0/0/0]ip address 10.0.34.4 24
[R4-GigabitEthernet0/0/0]int g0/0/2
[R4-GigabitEthernet0/0/2]ip add
[R4-GigabitEthernet0/0/2]ip address 10.0.45.4 24
[R4-GigabitEthernet0/0/2]quit
[R4]
```

- Initial configuration of R5

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R5
[R5]int LoopBack 0
[R5-LoopBack0]ip address 10.0.5.5 24
[R5-LoopBack0]int g0/0/2
[R5-GigabitEthernet0/0/2]ip address 10.0.45.5 24
[R5-GigabitEthernet0/0/2]quit
[R5]
```

3 Task Implementation

3.1 Checking Basic Configurations

Run the **display ip interface brief** command on R1, R2, R3, R4, and R5 to check interface configurations.

```
<R1>display ip int brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 3
The number of interface that is DOWN in Physical is 2
The number of interface that is UP in Protocol is 3
The number of interface that is DOWN in Protocol is 2
```

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	down	down
GigabitEthernet0/0/1	10.0.13.1/24	up	up
GigabitEthernet0/0/2	unassigned	down	down
LoopBack0	10.0.1.1/24	up	up(s)
NULL0	unassigned	up	up(s)

```
<R1>

[R2]display ip int brief
*down: administratively down
^down: standby
(l): loopback
```



(s): spoofing

The number of interface that is UP in Physical is 3

The number of interface that is DOWN in Physical is 2

The number of interface that is UP in Protocol is 3

The number of interface that is DOWN in Protocol is 2

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	down	down
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	10.0.23.2/24	up	up
LoopBack0	10.0.2.2/24	up	up(s)
NULL0	unassigned	up	up(s)

[R2]

[R3]display ip int brief

*down: administratively down

^down: standby

(l): loopback

(s): spoofing

The number of interface that is UP in Physical is 5

The number of interface that is DOWN in Physical is 0

The number of interface that is UP in Protocol is 5

The number of interface that is DOWN in Protocol is 0

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	10.0.34.3/24	up	up
GigabitEthernet0/0/1	10.0.13.3/24	up	up
GigabitEthernet0/0/2	10.0.23.3/24	up	up
LoopBack0	10.0.3.3/24	up	up(s)
NULL0	unassigned	up	up(s)

[R3]

<R4>display ip int brief

*down: administratively down

^down: standby

(l): loopback

(s): spoofing

The number of interface that is UP in Physical is 4

The number of interface that is DOWN in Physical is 1

The number of interface that is UP in Protocol is 4

The number of interface that is DOWN in Protocol is 1

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	10.0.34.4/24	up	up
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	10.0.45.4/24	up	up
LoopBack0	10.0.4.4/24	up	up(s)
NULL0	unassigned	up	up(s)

<R4>

<R5>display ip int brief



*down: administratively down

^down: standby

(l): loopback

(s): spoofing

The number of interface that is UP in Physical is 3

The number of interface that is DOWN in Physical is 2

The number of interface that is UP in Protocol is 3

The number of interface that is DOWN in Protocol is 2

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	down	down
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	10.0.45.5/24	up	up
LoopBack0	10.0.5.5/24	up	up(s)
NULL0	unassigned	up	up(s)

<R5>

Run the **ping** command on R3 and R5 to check the connectivity between devices.

[R3]ping 10.0.13.1

PING 10.0.13.1: 56 data bytes, press CTRL_C to break

Reply from 10.0.13.1: bytes=56 Sequence=1 ttl=255 time=130 ms

Reply from 10.0.13.1: bytes=56 Sequence=2 ttl=255 time=40 ms

Reply from 10.0.13.1: bytes=56 Sequence=3 ttl=255 time=60 ms

Reply from 10.0.13.1: bytes=56 Sequence=4 ttl=255 time=40 ms

Reply from 10.0.13.1: bytes=56 Sequence=5 ttl=255 time=50 ms

--- 10.0.13.1 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 40/64/130 ms

[R3]ping 10.0.23.2

PING 10.0.23.2: 56 data bytes, press CTRL_C to break

Reply from 10.0.23.2: bytes=56 Sequence=1 ttl=255 time=80 ms

Reply from 10.0.23.2: bytes=56 Sequence=2 ttl=255 time=20 ms

Reply from 10.0.23.2: bytes=56 Sequence=3 ttl=255 time=20 ms

Reply from 10.0.23.2: bytes=56 Sequence=4 ttl=255 time=20 ms

Reply from 10.0.23.2: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 10.0.23.2 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 20/34/80 ms

[R3]ping 10.0.34/4

Error: Unknown host 10.0.34/4.

[R3]ping 10.0.34.4

PING 10.0.34.4: 56 data bytes, press CTRL_C to break

Reply from 10.0.34.4: bytes=56 Sequence=1 ttl=255 time=50 ms



```
Reply from 10.0.34.4: bytes=56 Sequence=2 ttl=255 time=20 ms
Reply from 10.0.34.4: bytes=56 Sequence=3 ttl=255 time=40 ms
Reply from 10.0.34.4: bytes=56 Sequence=4 ttl=255 time=30 ms
Reply from 10.0.34.4: bytes=56 Sequence=5 ttl=255 time=30 ms
```

```
--- 10.0.34.4 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 20/34/50 ms
```

[R3]

[R5]ping 10.0.45.4

```
PING 10.0.45.4: 56 data bytes, press CTRL_C to break
Reply from 10.0.45.4: bytes=56 Sequence=1 ttl=255 time=90 ms
Reply from 10.0.45.4: bytes=56 Sequence=2 ttl=255 time=30 ms
Reply from 10.0.45.4: bytes=56 Sequence=3 ttl=255 time=20 ms
Reply from 10.0.45.4: bytes=56 Sequence=4 ttl=255 time=20 ms
Reply from 10.0.45.4: bytes=56 Sequence=5 ttl=255 time=30 ms
```

```
--- 10.0.45.4 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 20/38/90 ms
```

[R5]

3.2 Configuring Multi-area OSPF

Configure OSPF as planned. Manually specify the IP address of Loopback0 as the OSPF router ID on each device, and change the network type of Loopback0 to broadcast.

- Configure R1.

```
[R1] ospf 1 router-id 10.0.1.1
[R1-ospf-1] area 0.0.0.2
[R1-ospf-1-area-0.0.0.2] network 10.0.1.1 0.0.0.0
[R1-ospf-1-area-0.0.0.2] network 10.0.13.1 0.0.0.0
[R1-ospf-1-area-0.0.0.2] quit
[R1-ospf-1] quit
[R1] interface LoopBack0
[R1-LoopBack0] ospf network-type broadcast
```

- Configure R2.

```
[R2] ospf 1 router-id 10.0.2.2
[R2-ospf-1] area 0.0.0.3
[R2-ospf-1-area-0.0.0.3] network 10.0.23.2 0.0.0.0
[R2-ospf-1-area-0.0.0.3] quit
[R2-ospf-1] quit
[R2] interface LoopBack0
```



```
[R2-LoopBack0] ospf network-type broadcast
```

- Configure R3.

```
[R3] ospf 1 router-id 10.0.3.3
[R3-ospf-1] area 0.0.0.0
[R3-ospf-1-area-0.0.0.0] network 10.0.3.3 0.0.0.0
[R3-ospf-1-area-0.0.0.0] network 10.0.34.3 0.0.0.0
[R3-ospf-1-area-0.0.0.0] area 0.0.0.2
[R3-ospf-1-area-0.0.0.2] network 10.0.13.3 0.0.0.0
[R3-ospf-1-area-0.0.0.2] area 0.0.0.3
[R3-ospf-1-area-0.0.0.3] network 10.0.23.3 0.0.0.0
[R3-ospf-1-area-0.0.0.3] quit
[R3-ospf-1] quit
[R3] interface LoopBack0
[R3-LoopBack0] ospf network-type broadcast
```

- Configure R4.

```
[R4] ospf 1 router-id 10.0.4.4
[R4-ospf-1] area 0.0.0.0
[R4-ospf-1-area-0.0.0.0] network 10.0.4.4 0.0.0.0
[R4-ospf-1-area-0.0.0.0] network 10.0.34.4 0.0.0.0
[R4-ospf-1-area-0.0.0.0] area 0.0.0.1
[R4-ospf-1-area-0.0.0.1] network 10.0.45.4 0.0.0.0
[R4-ospf-1-area-0.0.0.1] quit
[R4-ospf-1] quit
[R4] interface LoopBack0
[R4-LoopBack0] ospf network-type broadcast
```

- Configure R5.

```
[R5] ospf 1 router-id 10.0.5.5
[R5-ospf-1] area 1
[R5-ospf-1-area-0.0.0.1] network 10.0.45.5 0.0.0.0
[R5-ospf-1-area-0.0.0.1] quit
[R5-ospf-1] quit
[R5] interface LoopBack0
[R5-LoopBack0] ospf network-type broadcast
```

3.3 Checking the Multi-Area OSPF Configuration

- Check the brief information about OSPF neighbor relationships on R3.

```
<R3>display ospf peer brief
```

```
OSPF Process 1 with Router ID 10.0.3.3
Peer Statistic Information
```

Area Id	Interface	Neighbor id	State
0.0.0.0	GigabitEthernet0/0/0	10.0.4.4	Full
0.0.0.2	GigabitEthernet0/0/1	10.0.1.1	Full
0.0.0.3	GigabitEthernet0/0/2	10.0.2.2	Full

```
<R3>
```

- Check the brief information about OSPF neighbor relationships on R5.



```
<R5>display ospf peer brief
```

```
OSPF Process 1 with Router ID 10.0.5.5
Peer Statistic Information
```

Area Id	Interface	Neighbor id	State
0.0.0.1	GigabitEthernet0/0/2	10.0.4.4	Full

```
<R5>
```

The command outputs show that the OSPF neighbor relationships between any two devices are normal.

- Check the OSPF routing table on R3.

```
<R3>display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.3.3
Routing Tables
```

```
Routing for Network
```

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.3.0/24	0	Stub	10.0.3.3	10.0.3.3	0.0.0.0
10.0.13.0/24	1	Transit	10.0.13.3	10.0.3.3	0.0.0.2
10.0.23.0/24	1	Transit	10.0.23.3	10.0.3.3	0.0.0.3
10.0.34.0/24	1	Transit	10.0.34.3	10.0.3.3	0.0.0.0
10.0.1.0/24	1	Stub	10.0.13.1	10.0.1.1	0.0.0.2
10.0.4.0/24	1	Stub	10.0.34.4	10.0.4.4	0.0.0.0
10.0.45.0/24	2	Inter-area	10.0.34.4	10.0.4.4	0.0.0.0

```
Total Nets: 7
```

```
Intra Area: 6 Inter Area: 1 ASE: 0 NSSA: 0
```

```
<R3>
```

R3 has learned the routes to all interfaces except R2's Loopback0 and R5's Loopback0 because the two Loopback0 interfaces do not have OSPF activated.

3.4 Importing External Routes to the OSPF Routing Table

- Configure R5 to import the Loopback0 interface route to its OSPF routing table.

```
[R5]ospf 1
[R5-ospf-1]import-route direct
[R5-ospf-1]
```

- Configure a default route on R2.

Configure a default route on R2, with Loopback0 specified as the outbound interface. Configure R2 to import the default route to its OSPF routing table, with the type of the external route being set to Type 1, cost being set to 20, and the **always** parameter not specified.

```
[R2] ip route-static 0.0.0.0 0.0.0.0 LoopBack 0
[R2]ospf 1
[R2-ospf-1]default-route-advertise type 1 cost 20
[R2-ospf-1]
```



- Check the imported external routes on R3 and test their connectivity.

```
<R3>display ospf routing 0.0.0.0
```

```
OSPF Process 1 with Router ID 10.0.3.3
```

```
Destination : 0.0.0.0/0
```

```
AdverRouter : 10.0.2.2
```

```
Tag : 1
```

```
Cost : 21
```

```
Type : Type1
```

```
NextHop : 10.0.23.2
```

```
Interface : GigabitEthernet0/0/2
```

```
Priority : Low
```

```
Age : 00h01m21s
```

```
<R3>
```

```
<R3>display ospf routing 0.0.0.0
```

```
OSPF Process 1 with Router ID 10.0.3.3
```

```
Destination : 0.0.0.0/0
```

```
AdverRouter : 10.0.2.2
```

```
Tag : 1
```

```
Cost : 21
```

```
Type : Type1
```

```
NextHop : 10.0.23.2
```

```
Interface : GigabitEthernet0/0/2
```

```
Priority : Low
```

```
Age : 00h01m21s
```

```
<R3>
```

```
<R3>display ospf routing 10.0.5.5
```

```
OSPF Process 1 with Router ID 10.0.3.3
```

```
Destination : 10.0.5.0/24
```

```
AdverRouter : 10.0.5.5
```

```
Tag : 1
```

```
Cost : 1
```

```
Type : Type2
```

```
NextHop : 10.0.34.4
```

```
Interface : GigabitEthernet0/0/0
```

```
Priority : Low
```

```
Age : 00h03m36s
```

```
<R3>
```

```
<R3>ping -c 2 10.0.5.5
```

```
PING 10.0.5.5: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.5.5: bytes=56 Sequence=1 ttl=254 time=50 ms
```

```
Reply from 10.0.5.5: bytes=56 Sequence=2 ttl=254 time=40 ms
```

```
--- 10.0.5.5 ping statistics ---
```

```
2 packet(s) transmitted
```

```
2 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 40/45/50 ms
```

```
<R3>ping -c 2 10.0.2.2
```

```
PING 10.0.2.2: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.2.2: bytes=56 Sequence=1 ttl=255 time=40 ms
```

```
Reply from 10.0.2.2: bytes=56 Sequence=2 ttl=255 time=30 ms
```

```
--- 10.0.2.2 ping statistics ---
```

```
2 packet(s) transmitted
```



```
2 packet(s) received
0.00% packet loss
round-trip min/avg/max = 30/35/40 ms
```

```
<R3>
```

3.5 Configuring Area 2 as a Stub Area

- Check the OSPF routing table on R1.

```
<R1>display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.1.1
Routing Tables
```

```
Routing for Network
```

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.1.0/24	0	Stub	10.0.1.1	10.0.1.1	0.0.0.2
10.0.13.0/24	1	Transit	10.0.13.1	10.0.1.1	0.0.0.2
10.0.3.0/24	1	Inter-area	10.0.13.3	10.0.3.3	0.0.0.2
10.0.4.0/24	2	Inter-area	10.0.13.3	10.0.3.3	0.0.0.2
10.0.23.0/24	2	Inter-area	10.0.13.3	10.0.3.3	0.0.0.2
10.0.34.0/24	2	Inter-area	10.0.13.3	10.0.3.3	0.0.0.2
10.0.45.0/24	3	Inter-area	10.0.13.3	10.0.3.3	0.0.0.2

```
Routing for ASEs
```

Destination	Cost	Type	Tag	NextHop	AdvRouter
0.0.0.0/0	22	Type1	1	10.0.13.3	10.0.2.2
10.0.5.0/24	1	Type2	1	10.0.13.3	10.0.5.5

```
Total Nets: 9
```

```
Intra Area: 2  Inter Area: 5  ASE: 2  NSSA: 0
```

```
<R1>
```

The command output shows that the default route is an OSPF external route.

- Configure area 2 as a stub area on R1 and R3.

```
[R1]ospf 1
[R1-ospf-1]area 2
[R1-ospf-1-area-0.0.0.2]stub
```

```
[R3]ospf 1
[R3-ospf-1]area 2
[R3-ospf-1-area-0.0.0.2]stub
```

- Check the OSPF routing table on R1 again.

```
[R1]display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.1.1
Routing Tables
```

```
Routing for Network
```



Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.1.0/24	0	Stub	10.0.1.1	10.0.1.1	0.0.0.2
10.0.13.0/24	1	Transit	10.0.13.1	10.0.1.1	0.0.0.2
0.0.0.0/0	2	Inter-area	10.0.13.3	10.0.3.3	0.0.0.2
10.0.3.0/24	1	Inter-area	10.0.13.3	10.0.3.3	0.0.0.2
10.0.4.0/24	2	Inter-area	10.0.13.3	10.0.3.3	0.0.0.2
10.0.23.0/24	2	Inter-area	10.0.13.3	10.0.3.3	0.0.0.2
10.0.34.0/24	2	Inter-area	10.0.13.3	10.0.3.3	0.0.0.2
10.0.45.0/24	3	Inter-area	10.0.13.3	10.0.3.3	0.0.0.2

Total Nets: 8

Intra Area: 2 Inter Area: 6 ASE: 0 NSSA: 0

[R1]

In this case, R1 does not have any OSPF external route. The original OSPF external routes 0.0.0.0/0 and 10.0.5.0/24 have been replaced by a default OSPF inter-area route.

- Check the OSPF LSDB on R1.

[R1]display ospf lsdb

OSPF Process 1 with Router ID 10.0.1.1
Link State Database

Area: 0.0.0.2						
Type	LinkState ID	AdvRouter	Age	Len	Sequence	Metric
Router	10.0.3.3	10.0.3.3	104	36	80000005	1
Router	10.0.1.1	10.0.1.1	109	48	80000006	0
Network	10.0.13.3	10.0.3.3	104	32	80000002	0
Sum-Net	0.0.0.0	10.0.3.3	152	28	80000001	1
Sum-Net	10.0.34.0	10.0.3.3	152	28	80000001	1
Sum-Net	10.0.3.0	10.0.3.3	152	28	80000001	0
Sum-Net	10.0.4.0	10.0.3.3	152	28	80000001	1
Sum-Net	10.0.45.0	10.0.3.3	152	28	80000001	2
Sum-Net	10.0.23.0	10.0.3.3	152	28	80000001	1

[R1]

R1 does not have Type 4 or Type 5 LSAs. The default route carried in the Type 3 LSA generated by the ABR is used to reach a destination outside the OSPF domain. In addition, Type 3 LSAs destined for other areas still exist.

This proves that an ABR blocks the transmission of Type 4 and Type 5 LSAs to the area that has been configured as a stub area and instead floods a default route destined for the ABR itself in this area through a Type 3 LSA.

3.6 Configuring Area 2 as a Totally Stub Area

- Configure area 2 as a totally stub area on R3.

[R3] ospf 1

[R3-ospf-1] area 0.0.0.2

[R3-ospf-1-area-0.0.0.2] stub no-summary

- Check the OSPF routing table and LSDB on R1 again.

[R1]display ospf routing



OSPF Process 1 with Router ID 10.0.1.1

Routing Tables

Routing for Network

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.1.0/24	0	Stub	10.0.1.1	10.0.1.1	0.0.0.2
10.0.13.0/24	1	Transit	10.0.13.1	10.0.1.1	0.0.0.2
0.0.0.0/0	2	Inter-area	10.0.13.3	10.0.3.3	0.0.0.2

Total Nets: 3

Intra Area: 2 Inter Area: 1 ASE: 0 NSSA: 0

[R1]

The original multiple OSPF inter-area routes have been replaced with only one default route 0.0.0.0/0, and the LSDB contains only one Type 3 LSA 0.0.0.0.

This proves that the ABR in a totally stub area blocks Type 3, Type 4, and Type 5 LSAs and instead generates a Type 3 LSA to advertise a default route destined for the ABR itself.

3.7 Configuring Area 1 as an NSSA

- Check the OSPF routing table on R4.

<R4>display ospf routing

OSPF Process 1 with Router ID 10.0.4.4

Routing Tables

Routing for Network

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.4.0/24	0	Stub	10.0.4.4	10.0.4.4	0.0.0.0
10.0.34.0/24	1	Transit	10.0.34.4	10.0.4.4	0.0.0.0
10.0.45.0/24	1	Transit	10.0.45.4	10.0.4.4	0.0.0.1
10.0.1.0/24	2	Inter-area	10.0.34.3	10.0.3.3	0.0.0.0
10.0.3.0/24	1	Stub	10.0.34.3	10.0.3.3	0.0.0.0
10.0.13.0/24	2	Inter-area	10.0.34.3	10.0.3.3	0.0.0.0
10.0.23.0/24	2	Inter-area	10.0.34.3	10.0.3.3	0.0.0.0

Routing for ASEs

Destination	Cost	Type	Tag	NextHop	AdvRouter
0.0.0.0/0	22	Type1	1	10.0.34.3	10.0.2.2
10.0.5.0/24	1	Type2	1	10.0.45.5	10.0.5.5

Total Nets: 9

Intra Area: 4 Inter Area: 3 ASE: 2 NSSA: 0

<R4>

The command output shows that R5 has an external route 10.0.5.0/24 described by a Type 5 LSA.



- Check the OSPF routing table on R5.

```
<R5>display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.5.5
Routing Tables
```

```
Routing for Network
```

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.45.0/24	1	Transit	10.0.45.5	10.0.5.5	0.0.0.1
10.0.1.0/24	3	Inter-area	10.0.45.4	10.0.4.4	0.0.0.1
10.0.3.0/24	2	Inter-area	10.0.45.4	10.0.4.4	0.0.0.1
10.0.4.0/24	1	Inter-area	10.0.45.4	10.0.4.4	0.0.0.1
10.0.13.0/24	3	Inter-area	10.0.45.4	10.0.4.4	0.0.0.1
10.0.23.0/24	3	Inter-area	10.0.45.4	10.0.4.4	0.0.0.1
10.0.34.0/24	2	Inter-area	10.0.45.4	10.0.4.4	0.0.0.1

```
Routing for ASEs
```

Destination	Cost	Type	Tag	NextHop	AdvRouter
0.0.0.0/0	23	Type1	1	10.0.45.4	10.0.2.2

```
Total Nets: 8
```

```
Intra Area: 1 Inter Area: 6 ASE: 1 NSSA: 0
```

```
<R5>
```

The default route in the OSPF routing table of R5 is described by a Type 5 LSA, which is generated by R2.

- Configure area 1 as an NSSA on R4 and R5.

```
[R4]ospf 1
```

```
[R4-ospf-1] area 0.0.0.1
```

```
[R4-ospf-1-area-0.0.0.1] nssa
```

```
[R5]ospf 1
```

```
[R5-ospf-1] area 0.0.0.1
```

```
[R5-ospf-1-area-0.0.0.1] nssa
```

- Check the OSPF routing table on R5 again.

```
<R5>display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.5.5
Routing Tables
```

```
Routing for Network
```

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.45.0/24	1	Transit	10.0.45.5	10.0.5.5	0.0.0.1
10.0.1.0/24	3	Inter-area	10.0.45.4	10.0.4.4	0.0.0.1
10.0.3.0/24	2	Inter-area	10.0.45.4	10.0.4.4	0.0.0.1
10.0.4.0/24	1	Inter-area	10.0.45.4	10.0.4.4	0.0.0.1
10.0.13.0/24	3	Inter-area	10.0.45.4	10.0.4.4	0.0.0.1
10.0.23.0/24	3	Inter-area	10.0.45.4	10.0.4.4	0.0.0.1
10.0.34.0/24	2	Inter-area	10.0.45.4	10.0.4.4	0.0.0.1



Routing for NSSAs

Destination	Cost	Type	Tag	NextHop	AdvRouter
0.0.0.0/0	1	Type2	1	10.0.45.4	10.0.4.4

Total Nets: 8

Intra Area: 1 Inter Area: 6 ASE: 0 NSSA: 1

The command output shows that there is no default route advertised by R2. Instead, there is an OSPF default route described by a Type 7 LSA, which is advertised by R4.

- Check the LSDB on R5.

[R5]display ospf lsdb

OSPF Process 1 with Router ID 10.0.5.5

Link State Database

Area: 0.0.0.1						
Type	LinkState ID	AdvRouter	Age	Len	Sequence	Metric
Router	10.0.5.5	10.0.5.5	79	36	80000005	1
Router	10.0.4.4	10.0.4.4	81	36	80000004	1
Network	10.0.45.5	10.0.5.5	79	32	80000001	0
Sum-Net	10.0.34.0	10.0.4.4	122	28	80000001	1
Sum-Net	10.0.13.0	10.0.4.4	122	28	80000001	2
Sum-Net	10.0.3.0	10.0.4.4	122	28	80000001	1
Sum-Net	10.0.1.0	10.0.4.4	122	28	80000001	2
Sum-Net	10.0.4.0	10.0.4.4	122	28	80000001	0
Sum-Net	10.0.23.0	10.0.4.4	122	28	80000001	2
NSSA	10.0.5.0	10.0.5.5	138	36	80000001	1
NSSA	10.0.45.0	10.0.5.5	138	36	80000002	1
NSSA	0.0.0.0	10.0.4.4	122	36	80000001	1

[R5]

The command output shows no Type 4 or Type 5 LSAs. Instead, external routes exist in the form of Type 7 LSAs (NSSA-LSAs).

- Check the OSPF routing table on R4.

[R4]display ospf routing

OSPF Process 1 with Router ID 10.0.4.4

Routing Tables

Routing for Network

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.4.0/24	0	Stub	10.0.4.4	10.0.4.4	0.0.0.0
10.0.34.0/24	1	Transit	10.0.34.4	10.0.4.4	0.0.0.0
10.0.45.0/24	1	Transit	10.0.45.4	10.0.4.4	0.0.0.1
10.0.1.0/24	2	Inter-area	10.0.34.3	10.0.3.3	0.0.0.0
10.0.3.0/24	1	Stub	10.0.34.3	10.0.3.3	0.0.0.0
10.0.13.0/24	2	Inter-area	10.0.34.3	10.0.3.3	0.0.0.0
10.0.23.0/24	2	Inter-area	10.0.34.3	10.0.3.3	0.0.0.0

Routing for ASEs

Destination	Cost	Type	Tag	NextHop	AdvRouter
-------------	------	------	-----	---------	-----------



```
0.0.0.0/0          22          Type1          1          10.0.34.3          10.0.2.2

Routing for NSSAs
Destination        Cost          Type          Tag          NextHop        AdvRouter
10.0.5.0/24        1            Type2          1            10.0.45.5      10.0.5.5

Total Nets: 9
Intra Area: 4   Inter Area: 3   ASE: 1   NSSA: 1

[R4]
[R4]display ospf routing
```

The external route 10.0.5.0/24 imported by R5 is described by a Type 7 LSA.

This proves that the NSSA blocks external Type 4 and Type 5 LSAs from being transmitted to this area and the ABR delivers a default route described by a Type 7 LSA to the NSSA. The ASBR delivers Type 7 LSAs to the NSSA to describe the AS external routes imported to this area.

4 Verification

Verification has been performed during the lab.

5 Quiz

1. (Multiple-answer question) Which of the following special areas are defined by OSPF?
(ABCD)
 - A. Stub Area
 - B. Totally Stub Area
 - C. Not-So-Stubby Area (NSSA)
 - D. Totally NSSA
2. (Question) What are the differences between a stub area and a totally stub area?
A stub area does not allow Type 4 or Type 5 LSAs to enter, but allows Type 3 LSAs to enter.
A totally stub area does not allow Type 4, Type 5, or Type 3 LSAs to enter, but allows only Type 3 LSAs that indicate the default route to enter.
3. (Question) On which router is inter-area route summarization configured?
ABR